

Yosra OMRANE
Junior AI Software Developer
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PROFESSIONAL SUMMARY

I'm a Junior AI Engineer passionate about building intelligent, real-world solutions. Experience in agent-based systems, automation workflows, and AI-driven tools at Talan Tunisia. Hands-on experience across end-to-end AI development and cross-functional team collaboration. I'm eager to grow with innovative teams and help bring production-ready AI solutions to life.

EDUCATION

Higher Institute of Computer Science and Mathematics Software Engineering	Monastir, Tunisia Sept. 2022 – July 2025
Higher Institute of Computer Science and Mathematics Integrated Preparatory Cycle	Monastir, Tunisia Sept. 2020 – May 2022

EXPERIENCE

IOPn (Freelance) AI Software Engineer	Remote Oct. 2025 – Present
– Neo-Council:	
- Architected and delivered a full-stack AI and Web3 governance platform enabling end-to-end crypto decision-making, from multi-agent analysis to secure on-chain execution. - Designed a scalable multi-agent architecture (risk, strategy, security, compliance, execution) orchestrating structured reasoning workflows and consensus-driven outputs. - Implemented human-in-the-loop approval systems with decision packets combining recommendations, risk scoring, confidence levels, and execution strategies. - Engineered a production-grade execution pipeline with transaction simulation, approval gating, wallet signing, and verifiable on-chain confirmation. - Built secure wallet-first authentication (SIWE + JWT), session-based collaboration layers, and a dynamic agent marketplace. - Ensured full auditability through a hash-chained, append-only event logging system. Technologies: Python, FastAPI, LangChain, RAG, FAISS, React, TypeScript, Solidity, Web3.py, MongoDB, Redis, Docker.	
– Neo-Investment:	
- Designed a multi-agent AI system coordinating technical, sentiment, on-chain, and regulatory analysis to generate high-confidence investment strategies. - Built a compliance-aware RAG pipeline with semantic retrieval and jurisdiction-based validation to enforce risk-aware recommendations. - Engineered a real-time intelligence pipeline integrating live market streams, news analytics, and proactive risk alerting. - Developed a quantitative risk engine with Monte Carlo simulations, VaR, Kelly sizing, and portfolio optimization metrics. - Implemented signal validation and backtesting frameworks to transform raw signals into statistically grounded insights. Technologies: Python, FastAPI, FAISS, Pandas, NumPy, SciPy, PyPortfolioOpt, React, TypeScript, Docker.	
– Neo-Defense:	
- Developed an AI-driven browser wallet assistant enabling natural-language transaction execution with automated and secure form handling. - Built a pre-execution security pipeline performing transaction simulation, phishing detection, and smart contract risk assessment. - Designed an intent-to-transaction engine converting user inputs into deterministic, auditable transaction representations. - Integrated advanced compliance checks, contextual safeguards, and advisory-only execution flows to ensure secure user interactions. - Implemented robust ML lifecycle pipelines including training, evaluation, monitoring, and controlled deployment strategies. Technologies: React, TypeScript, Node.js, FastAPI, PyTorch, Transformers, FAISS, Docker, GitHub Actions.	
– Apex Prediction:	
- Built an end-to-end forecasting pipeline covering data ingestion, feature engineering, model training, evaluation, and deployment. - Implemented hybrid forecasting models (LightGBM, NHITS) with walk-forward validation and performance optimization. - Delivered advanced explainability using SHAP and permutation importance for both global and local prediction insights. - Designed an interactive analytics platform with scenario simulation and AI-driven decision support (Buy/Sell/Hold). Technologies: Python, LightGBM, PyTorch, SHAP, Streamlit, Pandas, NumPy, FAISS.	
Talan Tunisie AI Software Engineer	Tunis, Tunisia July 2025 – Present
– Contribution to Technical & Innovation Projects:	

- Contributed to migrating JEE legacy systems to modern Spring Boot microservices architecture.
- Developed an Excel-integrated MCP-based prototype to explore AI assistance and workflow automation.
- Implemented n8n workflows for news distribution, including LinkedIn automation and sentiment analysis.
- **Technologies:** Python, MCP, Excel API, n8n, LinkedIn API, Sentiment Analysis, Git, GitLab, Java, Spring Boot.

– **MSAP Project & Cyttest Collaboration:**

- Implemented template matching in DefectDojo and built a productivity-focused Chrome extension for security analysts.
- Created an AI-assisted vulnerability reporting tool compliant with internal standards for detailed technical analysis.
- Deployed LLM models on AWS GPU instances using containerized pipelines and GitLab CI/CD.
- **Technologies:** Python, LLMs, RAG, GitLab, Docker, AWS EC2, GPU instances, DefectDojo, Chrome API.

– **Mentoring & Team Support:**

- Mentored 7 final-year students, providing technical guidance and structuring for four diverse AI projects.
- Led the BrandOrb team during Summer Camp to a 1st place (Gold Prize) victory through strategic project planning.
- **Skills & Tools:** Mentorship, Agile (Scrum/Kanban), Jira, Team Coordination, Strategic Planning.

Talan Tunisie

AI Software Engineering Intern

Tunis, Tunisia

Feb. 2025 – May 2025

- Developed a VS Code extension featuring a multi-agent software assistant using LangGraph and MCP servers.
- Conducted systematic benchmarking of LLMs to evaluate performance and latency in real-world development scenarios.
- **Technologies:** Python, LangGraph, MCP, TypeScript, VS Code API, LLMs.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript
Web Development Frameworks: React.js, Node.js, Flask, FastAPI
Databases: SQL (PostgreSQL), NoSQL (MongoDB), Knowledge Graph (Neo4j)
Vector Databases: FAISS, ChromaDB
AI & Machine Learning: TensorFlow, Keras, PyTorch, Scikit-learn, Hugging Face Transformers, LangGraph, MCP Servers, LangChain, RAG, Graph RAG, Agentic RAG, Agentic AI, LLMs, LlamaIndex, TTS
DevOps and Cloud: Docker, GPU Servers, CI/CD (GitHub Actions, GitLab CI), AWS
Tools and Collaboration: Git, GitHub, GitLab, Postman, Swagger
Software Testing: JUnit, Pytest
Development Environment and OS: VS Code, Google Colab, Kaggle, Lightning AI, Linux, Windows
Methodology and Tools: Agile (Scrum/Kanban), Trello, Jira

LANGUAGES

Arabic (Native) | **French** (Fluent) | **English** (Advanced)

CERTIFICATIONS

AWS Academy Cloud Architecting AWS Academy	2024
AWS Academy Cloud Security Foundations AWS Academy	2024
AWS Academy Cloud Foundations AWS Academy	2024